François Leroy

Macroecology, Statistical Modelling (ML, DL)

466 W 2nd Av.
Columbus, OH 43201, USA
+13802157939
francois.libert.leroy@gmail.com
https://github.com/FrsLry

Experience

2024-Ongoing **Postdoctoral researcher - Using Artifical Inteligence to understand spatio-temporal changes of biodiversity**, *The Ohio State University, dept. of Evolution, Ecology and Organismal Biology*, Columbus, Ohio.
- Creating neural hierarchical models from Bayesian framework
- Tailoring loss functions to infer distribution parameters
- Creating simulated data to test models
- Hired for the project ABC Global Center, collaborating with computer scientists from MIT, McGill University, Mila institute (Montreal, Canada)

2020-2024 **PhD. - Spatial scaling and decomposition of macroecological changes**, *Faculty of Environmental Sciences, CZU, dept. of Spatial sciences*, Prague.
- Using big ecological data to model biodiversity changes across spatial scales
- Using model optimisation and selection on various machine learning algorithms (Random Forest, BRT, XGBoost, linear models...)

Education

2021 (1 semester) **Machine Learning**, *Faculty of Mathematics and Physics, UFAL, Charles Univesity*, Prague.
Studying all Machine Learning algorithms, from Support Vector Machines to Neural Networks

2020-2024 **PhD. - Spatial scaling and decomposition of macroecological changes**, Prague.

2018-2020 **Marine Sciences MSc**, *Sorbonne University*, Paris.
Numerical Ecology, modelling, geostatistics, GIS, oceanography, marine ecology, biogeochemistry, database management

2015-2018 **Bachelor of Science**, *South Brittany University*, Vannes (France).
Specialized in Coastal Ecosystems and Management, GIS

Teaching

2021-2024
- Statistical ecology and macroecology
- Version Control using Git and Github
- GIS and spatial analysis

Modelling skills

Deep Learning Multilayer Perceptron, Convolutional/Recurrent/Hierarchical Neural Networks

Machine Learning Classification and Regression Tree based algorithms (RF, BRT, GBM, XGBoost), Linear Models (GLM, Mixed Models, polynomial regressions...), Hierarchical modelling

Others Generalized Additive Models, Bayesian inference with MCMC algorithms, Bayesian Networks, Hidden Markov Models, Time series analysis, Model: optimisation (e.g. regularization), prediction, scalability

Programming skills

Advanced: Python, R, Git, QGIS, ArcGIS, LaTeX

Intermediate: Shell, MySQL

Basic: Julia, MATLAB, HTML5, CSS

Highlighted Internships and others

2024 (1 semester) **Deep Learning**, *Faculty of Mathematics and Physics*, Charles University, Prague.
- Going through all Deep Learning algorithms

2022 (1 week) **HMSC**, *Jyväskylä summer school*, Jyväskylä , Finland.
- Summer school on Hirearchical Modeling of Species Community

2020 (6 months) **Community Modelling**, *DYNECO-LEBCO, IFREMER*, Brest (France).
- Bayesian networks and qualitative modelling to assess the impact of environmental changes on benthic communities

Highlighted Talks and Conferences

Invited speaker 2024-02-16 **Introduction to Reproducible Science: Version Control using Git and Github**, *Ecoinformatics IAVS*, Online, Slides.

Conference 2023-08-10 **Decomposing abundance change to recruitment and loss: analysis of the North-American avifauna**, *Ecological Society of America*, Portland, OR, Slides.

Conference 2022-06-05 **Untangling biodiversity changes across a continuum of spatial scales**, *International Biogeography Society conference*, Vancouver, BC, Slides.

Scientific Referees

Dr. **Marta Jarzyna**, The Ohio State University, ☎+1 (978) 587-5938, jarzyna.1@osu.edu

Dr. **Petr Keil**, Czech University of Life Sicences, ☎+420 224382659, keil@fzp.czu.cz

Dr. **Martin Marzloff**, Ifremer, ☎+332 98224327, Martin.Marzloff@ifremer.fr

Dr. **Vítězslav Moudrý**, Czech University of Life Sicences, ☎+420 224382653, moudry@fzp.czu.cz

References

[1] **François Leroy**, Marta A. Jarzyna, and Petr Keil. "Acceleration hotspots of North American birds' decline are associated with agriculture". en. *Science* 391.6788 (2026), pp. 917–921. DOI: 10.1126/science.ads0871. URL: https://www.science.org/doi/10.1126/science.ads0871.

[2] Carmen D. Soria, Gabriel R. Ortega-Solís, Friederike J. R. Wölke, Vojtěch Barták, **François Leroy**, Kateřina Tschernosterová, Vladimir Bejček, Sergi Herrando, Ivan Mikuláš, Karel Šťastný, Mutsuyuki Ueta, Petr Voříšek, and Petr Keil. "Spatial Autocorrelation of Species Diversity and Distributions in Time and Across Spatial Scales". en. *Global Ecology and Biogeography* 35.3 (2026), e70221. DOI: 10.1111/geb.70221. URL: https://onlinelibrary.wiley.com/doi/10.1111/geb.70221.

[3] Petr Keil, Adam T. Clark, Vojtěch Barták, and **François Leroy**. "Should Regional Species Loss Be Faster or Slower Than Local Loss? It Depends on Density-Dependent Rate of Death". en. *Ecology and Evolution* (2025). DOI: 10.1002/ece3.71162. URL: https://onlinelibrary.wiley.com/doi/10.1002/ece3.71162.

[4] **François Leroy**. "Spatial scaling and decomposition of macroecological changes". SUPERVISOR: Mgr. Petr Keil, Ph.D. Doctoral theses, Dissertations. Czech University of Life Sciences Prague, Faculty of Environmental SciencesPraha, 2024. URL: https://theses.cz/id/aigse5/.

[5] Dominika Prajzlerová, Vojtěch Barták, Petr Keil, Vítězslav Moudrý [...], **François Leroy** [...], and Petra Šímová. "The relationship between remotely-sensed spectral heterogeneity and bird diversity is modulated by landscape type". *International Journal of Applied Earth Observation and Geoinformation* (2024). DOI: 10.1016/j.jag.2024.103763. URL: https://linkinghub.elsevier.com/retrieve/pii/S1569843224001171.

[6] **François Leroy**, Jiři Reif, David Storch, and Petr Keil. "How has bird biodiversity changed over time? A review across spatio-temporal scales". *Basic and Applied Ecology* 69 (2023), pp. 26–38. DOI: 10.1016/j.baae.2023.03.004. URL: https://www.sciencedirect.com/science/article/pii/S1439179123000117.

[7] **François Leroy**, Jiří Reif, Zdeněk Vermouzek, Karel Šťastný, Eva Trávníčková, Vladimír Bejček, Ivan Mikuláš, and Petr Keil. "Decomposing biodiversity change to processes of extinction, colonization, and recurrence across scales". *Ecography* (2023). DOI: 10.1111/ecog.06995. URL: https://onlinelibrary.wiley.com/doi/10.1111/ecog.06995.

[8] Vítězslav Moudrý, Petr Keil, Anna F Cord [...], **François Leroy** [...], and Petra Šímová. "Scale mismatches between predictor and response variables in species distribution modelling: A review of practices for appropriate grain selection". *Progress in Physical Geography: Earth and Environment* (2023). DOI: 10.1177/03091333231156362. URL: https://doi.org/10.1177/03091333231156362.

[9] Vítězslav Moudrý, Kateřina Gdulová, Lukáš Gábor, Eliška Šárovcová, Vojtěch Barták, **François Leroy**, Olga Špatenková, Duccio Rocchini, and Jiří Prošek. "Effects of environmental conditions on ICESat-2 terrain and canopy heights retrievals in Central European mountains". *Remote Sensing of Environment* (2022). DOI: 10.1016/j.rse.2022.113112. URL: https://www.sciencedirect.com/science/article/pii/S0034425722002267.